

Physiological and Ergonomic Studies at Antarctica

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ABSTRACT

Physiological responses to understand effect of extreme condition, viz. sub zero temperature, low humidity and high wind velocity along with Ergonomic studies were carried out at base station, Maitri during 23rd summer expedition 2003 -2004. Recordings were made on 14 volunteers from scientific and logistic groups of the expedition team before the start of voyage, on-board ship to Antarctica and throughout the summer months at Maitri (70°45'S, 11°44'E). Physiological parameters like Heart Rate, Blood Pressure, Hand grip strength and Pedometer studies along with Ergonomic studies and Habitability survey were carried out. It was observed that heart rate increased from base-line value of 75 bpm $\pm$ 0.13 to 85 bpm $\pm$ 0.21 ($p < 0.001$), systolic blood pressure increased from 119.9 mm Hg $\pm$ 0.15 to 128.4 mm Hg $\pm$ 0.53 ($p < 0.001$) and diastolic Blood pressure increased from 79.9 mm Hg $\pm$ 0.14 (SEM) to 84.3 mm Hg $\pm$ 0.12 during the voyage ($p < 0.001$). The rise was persistent till the 4th week of stay. Heart Rate and Systolic Blood pressure showed a decline after the 4th week whereas Diastolic Blood Pressure decline was not significant. The recordings of physiological parameters did not reach pre-voyage (base-line) levels. Energy consumption increased from base-line value of 219 $\pm$ 0.95 kcal/hr to 297 $\pm$ 0.32 kcal/hr suggesting need for more energy during normal walk on Antarctic terrain as compared to level ground. No change was observed in Handgrip Strength of the volunteers during study. The over-wintering members did suggest the need of improvement for making their stay more comfortable at Maitri. Ergonomic study showed that immediate attention was needed towards electrical and flooring maintenance at the base station workshop.

INTRODUCTION

Sustaining life at polar station involves many stressful factors. Most of the studies carried out at different Antarctic stations have shown marked amount of variations in the physiological parameters of team members. Extensive studies have been conducted to evaluate the changes in body functions induced by various stress at Antarctica. Most researchers

have concentrated to study the state of individual function system in response to these environmental and psychological stresses (Budd & Warhaft, 1966; Wyndhan et al., 1964; Wyndhan et al. 1968; Duncan, 1988; Buguet et al., 1987). Subsequent studies by American, European and Australian researchers started exploring the performance changes as an effect of Antarctic exposure (Soroko et al, 1984; Palinkas 1987; Bodey, 1978). It was shown from studies conducted by Kruk et al and Bittel et al, that moderate cold may have minimal effect upon healthy individuals (Bittel et al, 1988; Kurk et al, 1991), whereas severe continued exposure has negative influence. Herr and White showed that due to extreme continuous exposure to cold, normal physiological balance of individuals get altered (Herr & White, 1991). The data available on impact of such extreme conditions likely extreme low temperature, high wind velocity and restricted mobility on physiological variables, are very limited in reference with tropical expedition members.

Present study was conducted on 23rd Indian Antarctic Expedition Members to elucidate a) the effects of Antarctic environment on physiological parameters b) requirement of energy expenditure during normal walking on Antarctic terrain and c) Ergonomic studies at Indian Antarctic Base Station, Maitri (70°45'S, 11°44'E) during 2003 – 2004.

MATERIALS AND METHODS

The observations were made before commencement of journey from Cape Town, on-board ship and during stay at Maitri (January 2004 to March 2004).

Subjects: Fourteen expedition members gave consent and volunteered for the study. They were aged between 24years and 54 years (38.5 ± 1.46 years). Their initial weight ranged from 52.48kgs to 78.56kgs (65.52 ± 1.49 kgs) and height between 166cms and 182cms (174 ± 1.12 cms). The volunteers were from scientific and logistic group.

a) *Physiological Parameters:* Heart Rate was monitored using Polar Heart rate monitor (Polar electro Model No. CE 0537, Malaysia) for 24 hours having regular intervals between readings of every subject before, while travelling and during stay at Maitri. Blood Pressure Monitor was monitored using Oscillation Method based Blood Pressure Monitor (Make: Omron CH 606, USA). Hand grip Dynamometer (Jammer, USA) was used to measure hand grip strength of each volunteers during travel and throughout the summer months at Antarctica.

Subsequent recordings of Systolic Blood Pressure, Diastolic Blood Pressure and Hand Grip Strength at Maitri were made on weekly basis.

b) Habitability Study: Questionnaire was provided to volunteers residing at Maitri (previous wintering team members) and also the visiting summer team members. They were asked to read the questionnaire carefully and respond to the set of questions candidly.

c) Ergonomic Studies: Ergonomic study of 'Gen Bhagat' workshop was carried out along with the 'Jan Gan Man' Generator Set Enclosure.

Statistical Analysis: The data are presented as mean $\pm$ SEM. Statistical analysis of the data was done using t-test. $P < 0.05$ was considered significant.

RESULTS

Heart Rate: base-line recording was taken just before sailing towards Antarctica, at Cape Town, South Africa. Subsequent readings were taken during voyage. The next readings were taken weekly during stay at Maitri. Fig. 1 shows that during journey there was steady rise in 24 hour heart rate. The rise persisted on reaching Antarctica. Further readings after staying at base station showed a decline in heart rate after 4th week but it never touched base-line.

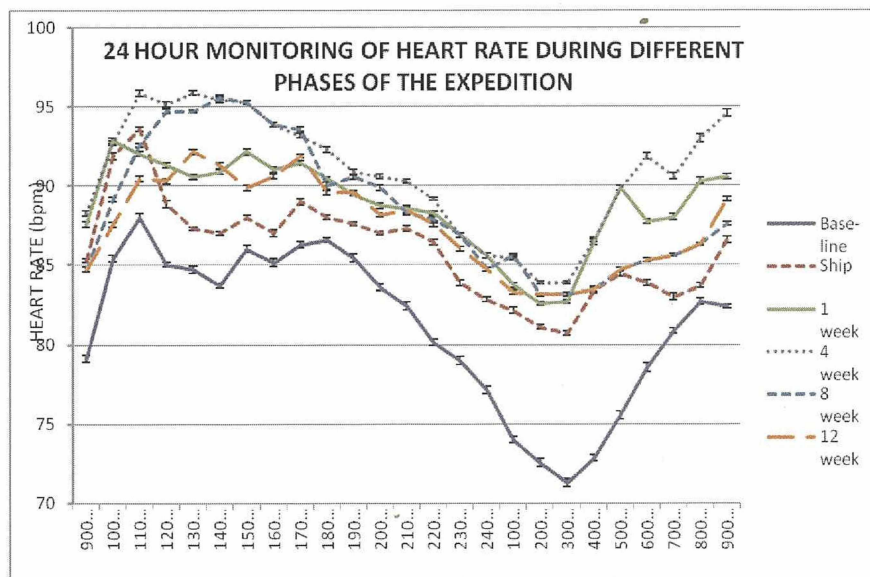


Fig. 1: 24 hour monitoring of heart rate during different stages of expedition. Heart rate increased during journey and showed a decline after stay at Antarctica for 4 weeks but did not reach normal base-line readings.

SYSTOLE AND DIASTOLE BLOOD PRESSURE

Both systolic and diastolic blood pressures showed a tendency to rise during voyage ($p<0.001$). The pattern of rise continued even on reaching base station. After staying at Maitri for 4 weeks systolic blood pressure showed a decline but did not reach base-line recording. No remarkable decline was noticed in diastolic blood pressure (Figs 2 & 3).

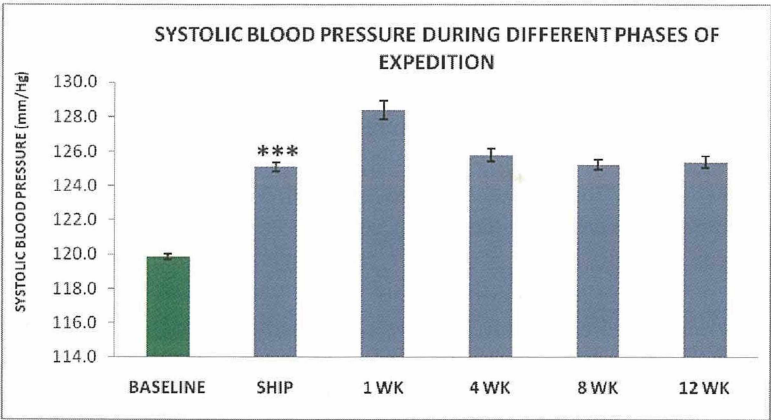


Fig. 2: Systolic blood pressure during different phases of expedition

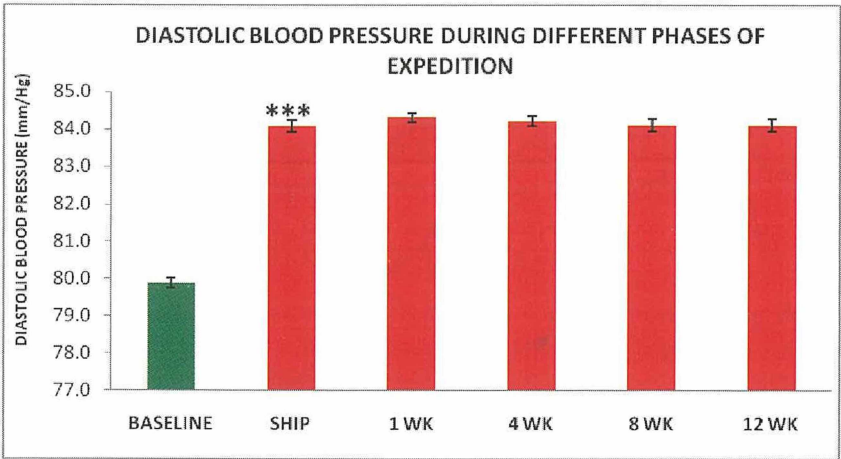


Fig. 3: Diastolic Blood Pressure during different phases of expedition

Hand Grip Strength: No significant change was noticed between the base-line and subsequent hand grip strength data. Average hand grip strength was found to be 48 ± 0.42 kgs.

Pedometer Studies:

Walking the same distance on Antarctic soil showed more energy consumption as compared to normal walking on level ground. There was significant increase of energy consumption ($p < 0.001$) during normal walking on Antarctic terrain. The average energy expenditure was found to be 297 ± 0.32 kcal / hour (Fig. 4).

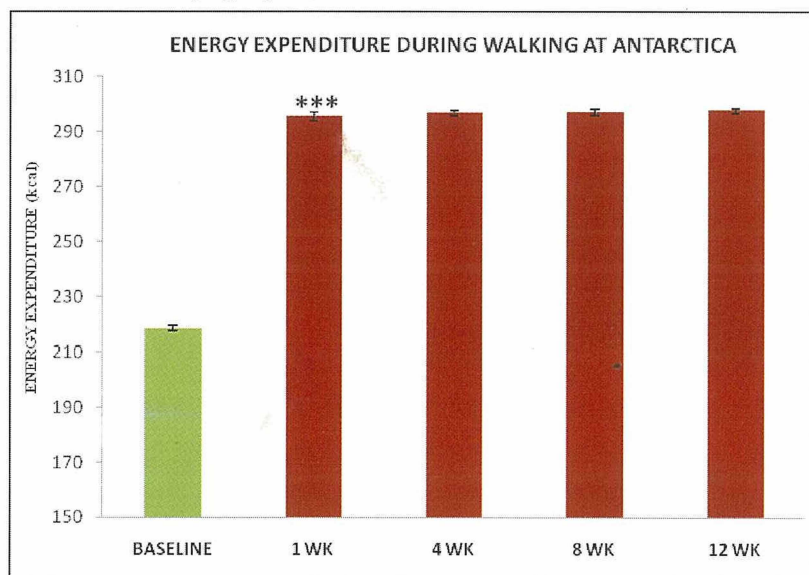


Fig. 4: Energy expenditure during walking at level ground compared with walking on Antarctic Soil

Habitability Studies:

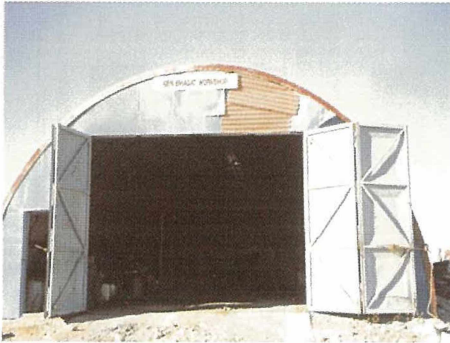
Questionnaire was handed over to the previous winter team members, willing to participate in the survey and visiting summer team members. They were asked to answer the questionnaire without biasness. From their response it was found that most expedition members (over-wintering) showed dissatisfaction about the toilet system at station. They were satisfied with the rooms provided but suggested the need for better recreational facilities at the station. 73.6% of the over-wintering members did complain of isolation and loneliness.

Ergonomic Studies:

Study was carried out at the 'Gen Bhagat' Workshop at Maitri. Studies on Generator set enclosure providing electricity to the station were also done.

ERGONOMIC OBSERVATIONS AND RECOMMENDATIONS

- a) Smooth flooring is needed for better mobility and workability.
- b) Proper illumination with transparent ceiling and sealed windows is required for better visibility and comfortable working atmosphere inside the workshop.



Pic 1: Gen Bhagat Workshop – Entrance

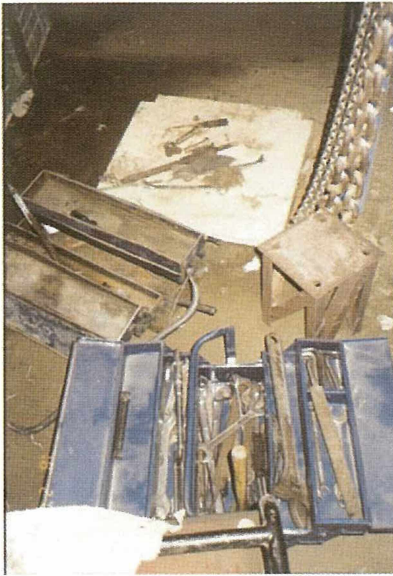


Pic 2: Floor of Gen Bhagat Workshop



Pic 3: Gen Bhagat Workshop – Oil Spillage on Working Ramp

- a) Strict vigil is needed to avoid oil spillage for preventing 'Slip-and-Fall' incidence.



(a)

Pic 4(a): Tool Box at Gen Bhagat Workshop,



(b)

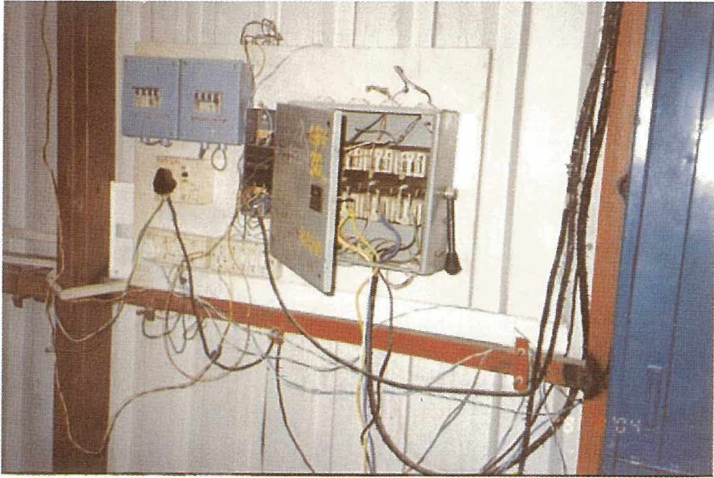
Pic 4(b): Stacking Rack for Hose at Gen Bhagat Workshop

- a) Tool box need to be maintained properly.
- b) The stacking rack need to be maintained properly and systematically.



Pic 5: 'Fire point' at Gen Bhagat work shop

- a) A proper 'Fire Point' is provided at the work shop. Proper checking of expiry dates and maintenance of the extinguishers should be strictly carried out.

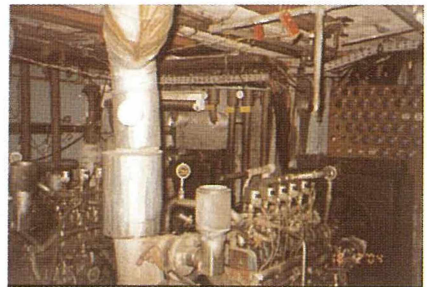


Pic 6: Electrical Junction Box at Gen Bhagat Workshop

- a) Proper dressings of naked wires to be done immediately.
- b) All wires need to be concealed and
- c) Properly insulated



(A)

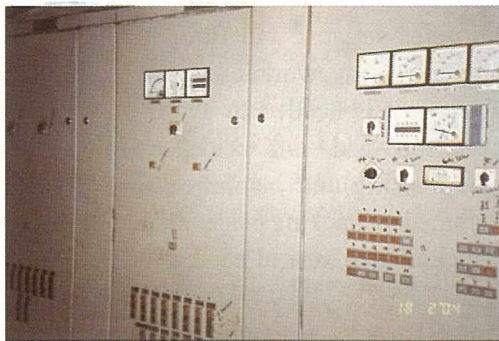


(B)

Pic 7: (A) 'Jana Gana Mana' Genset enclosure, (B) Inside view of enclosure

- a) Proper Illumination is needed inside the enclosure.

b) Transparent perplex sheet sealed windows can be used to provide illumination inside enclosure during day-light.



Pic 8: Control Panel of Genset

a) Proper Tagging / Labeling needs to be done on the controls.

DISCUSSION

The increase in physiological parameters started during the voyage and persisted even at base station. Systolic pressure showed a decline after 4 weeks of Antarctic stay, whereas Diastolic pressure did not show any significant decline. The subsequent heart rate and systolic blood pressure readings after 4 weeks showed a decline but did not reach base-line recordings. The rise initially may be due to stress of voyage, during the journey towards Antarctica. There were many who complained of sea sickness, nausea, vomiting and headache. On reaching Antarctica, exposure to low temperature aggravated the voyage stress. Both heart rate and blood pressure continued to be higher till 4th week of stay at Maitri. Previous study has also shown that physiological parameters along with autonomic responses started declining after 4 to 6 weeks of stay. This trend may be due to the acclimatization factor that to an extent attunes the members to Antarctic surrounding. The results of the present study are in accordance with previous studies (National Institute of Polar Research, 1979, Harinath et al 2004). It is known that for normal walking on level ground, 225kcal of energy is required per hour (Ainsworth, B.E.; 2002, The Compendium of Physical Activities Tracking Guide, and Pangrazi 2003). When pedometer studies were carried out at Antarctica, it was observed that the amount of energy required was more than normal. This may be effect of rugged terrain encountered at Antarctica which may also be the cause of restricted

movement of members around base camp resulting diminished physical fitness of over-wintering members.

Ergonomic studies suggest that floor of the workshop should be smoothened for proper walking, workability and mobility of vehicles. Apart from flooring, strict vigil is needed to avoid oil spillage to prevent the probability and risk of 'slip – and – fall' incidence while working. The electrical wires need proper dressing and insulation to avoid electrocution and malfunction of equipments. Stacking places should be given proper attention for making things available in time of need.

CONCLUSION

It may be concluded that moving over Antarctica soil demands more energy per hour compared to walking on level ground and sympathetic activity increased throughout the period of stay at Antarctica.

RECOMMENDATIONS

Over – wintering members suggested need for better basic amenities and recreational facilities for the period of winter months. Proper care should be taken for maintaining the generator rooms and workshop, providing better illumination and flooring respectively, to prevent any untoward incidence.

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